

1. PREPRATORY WORK BEFORE COMMENCEMENT OF EXTRACTION

1.1 Extraction of coal from the panel shall not be commenced till 6th Chimney is connected to proposed -92.5 MRL lateral.

1.4 Workings within 60m of the face in the same seam or any other seam shall be kept free from water before and during the extraction of Panel LT-3C & LT-3D.

3. MANNER OF EXTRACTION

3.1 The manner of extraction shall be by Jankowice method, so however that the maximum span from stowed sand to the face at any stage shall not exceed 5.5m.

3.2 Not more than 2.4m (true) thickness of the seam shall be extracted in 1st, 2nd and 3rd lifts. A parting of 0.6m thick coal with underlying lift shall be maintained.

3.3 The faces shall be advanced at an uniform rate and extraction of immediate overlying lift shall not commence unless the bottom lift has advanced by 20m ahead.

3.4 Total length of the panel along strike shall be worked block wise and by breasting. Block along the strike shall be made by driving staples of length not more than 5.5m measured from barricade and the width of breasting not exceeding 4.5m. Block shall not be more than 30m in length.

3.5 After extracting each block, the void created shall be stowed up-to 4.5m from the stowed goaf leaving 1m clearance between the face and barricade. Extraction in the next block shall be started only after adjoining voids has been stowed completely.

3.6 Each staple shall be breasted by putting shot holes so placed that there is minimum damage to the support in the area along strike.

3.7 Through staple upper lifts shall be opened only on the stowed goaf of lower lift.

3.9 Each rise/cut shall be stowed immediately after its use is over.

3.10 Flumes of adequate size and the suitable material shall be installed at suitable places for proper drainage of stowing water.

3.11 Barricade of adequate strength to contain the sand shall be made.

3.12 The rate of extraction shall be as uniform as practicable and one cycle of operations (including stowing) shall be completed within 36 hours.

3.13 The extraction in each lift shall be completed within a period of 9 (nine) months. The panel shall be isolated by isolation stoppings after completion of extraction.

4. GROUND MOVEMENT MONITORING

4.1 Convergence recorders, Load cells and bore hole extensometers shall be installed at 10m interval at both the gate roadways/chimney, 1st and 2nd Horizon below the permitted panel. Records of strata control monitoring scheme along with interpretation shall be submitted to the DGMS quarterly.

5. VENTILATION CONTROL

5.1 Ventilation of the face shall be done as per the statute and shall be ventilated by two intakes (two cuts) and two returns (two cuts). 1000 m³ of air per minute shall always be kept available in the production face.

5.3 (a) In order to detect spontaneous heating at early stage, the air in the return airway of the longwall face shall be tested for percentage of carbon monoxide once atleast every day by Assistant Manager/Ventilation Officer with an approved automatic detector. The air in the return airway shall be completely analysed once atleast in every 7 days with a view to determine the ratio of carbon monoxide formed/ oxygen absorbed.

(b) On observation of alarming ratio, workings in the panel shall be stopped forthwith under intimation to the DGMS.

(c) Thermo-compositional investigation shall be carried out on regular basis for early detection of spontaneous heating during extraction by engaging any scientific agency.

5.4 The environmental parameters in the panel like methane and carbon monoxide shall be constantly monitored through suitable instrumentation. An automatic device with audio-visual alarm for monitoring of inflammable gas in return airway of the district shall be kept installed. Such device shall be electronically interlocked to the main power switch of the district in such a way that the electric power cuts off in the district/panel as soon as percentage of CH₄ exceeds 0.5% in the return airway.

2. SUPPORT AT FACES

(a) Roof shall be supported by timber cross-bars on five rows of props. First row of props shall be set close to the face at a distance of 0.35m from the face.

(b) Props shall be set at a maximum interval of 0.8m between the props in the same row and also at 0.8m between the rows. Props shall be set vertically in the notch (15cm deep) in the floor.

(c) The length of the cross-bar shall not be less than 2.6m. The cross-bar must be adjusted in such a way that no roof is left unsupported in between the barricade and the face.

(d) All faults, visible slips, breaks in the roof shall be supported by cogs at an interval of 2.4m on either side of such disturbance.

(e) Steel/wooden chock/cog shall be set at a distance of 2.4m from the from the stowed sand and also at 2.4m interval along the strike when the span from stowed sand to the face exceeds 3.6m.

3. SUPPORT OF GATE ROAD/RISE

3.1 The gate roads/ chimney shall be kept supported systematically by steel or wooden cross bars at 1m interval with timber laggings.

Or

3.2 The gate roads/ chimney shall be kept supported with full column grouted roof bolts of 1.5m length and 22mm diameter made of ribbed bar in a square grid of 1.2m.

3.3 All the roof bolts shall be perpendicular to the bedding plane. The roof bolts shall be tightened with 15cm x 15cm x 6mm plate and nuts.

4. SUPPORT AT LEDGES/OVER HANGS:

- a) All ledges in the roof shall be kept supported by cogs.
- b) All over hangs which cannot be dressed down shall be supported by the stay props at a maximum of the two suitable horizons. The lid at the top of each such prop shall not be less than 1.0m in length.
- c) All faults, visible slips, breaks in the roof shall be supported by cogs at an interval of 2.4m on either side of such disturbance.

5. The lid and wedges used with the props shall have width not less than diameter of the prop, thickness not less than 8 cm. and a length not less than 0.5m.

6. Cross-bar shall be supported on prop. Every cross-bar shall be made tight against the roof. If the lagging is necessary, in case of uneven roof, for the purpose, the number of laggings used on the cross-bar shall not be less than one for every one meter length of bar and the lagging shall be kept tight.

7. SUPPORT BY PACK WALL OR SAND:

(a) The stowing shall be so advanced that at no time the distance between the face and stowing is more than 5.5m.

(b) Maximum span in the case of long wall is permitted to 5.5m from the barricade props.

8. ADDITIONAL SUPPORT SHALL BE REECTED AS AND WHEN NECESSARY:

While driving a stable the shorter cross-bars may be removed only after erecting longer cross bar. Where roof is not sound the shorter cross bars should not be removed and it should remain in position in addition to longer cross bars.